

Math 242, S15

Quiz 2

Name: Answer

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Find a general solution (explicit if convenient) for the first-order differential equations $\frac{dy}{dx} = 2x \csc(y)$.

$$\frac{dy}{dx} = 2x \csc(y)$$

$$\frac{1}{\csc(y)} dy = 2x dx$$

$$\sin y dy = 2x dx \Rightarrow \int \sin y dy = \int 2x dx$$

$$-\cos y = x^2 + C$$

$$\cos y = -x^2 - C$$

$$y(x) = \cos^{-1}(-x^2 - C)$$

Problem 2. (5 points) Find the explicit particular solution for the initial value problem

$$\frac{dy}{dx} = 4x^3 y - y, \quad y(1) = -4.$$

$$\frac{dy}{dx} = 4x^3 y - y \Rightarrow \frac{dy}{dx} = y(4x^3 - 1)$$

$$\frac{1}{y} dy = (4x^3 - 1) dx$$

$$\int \frac{1}{y} dy = \int 4x^3 - 1 dx$$

$$\ln|y| = x^4 - x + C$$

$$|y| = e^{x^4 - x + C} \Rightarrow |y| = A e^{x^4 - x} \quad (A > 0)$$

$$\Rightarrow y = A e^{x^4 - x} \text{ or } y = -A e^{x^4 - x}$$

$$y(1) = -4 \Rightarrow y = -A e^{x^4 - x} \text{ and } -A e^0 = -4$$

$$A = 4$$

$$\Rightarrow y(x) = -4 e^{x^4 - x}$$